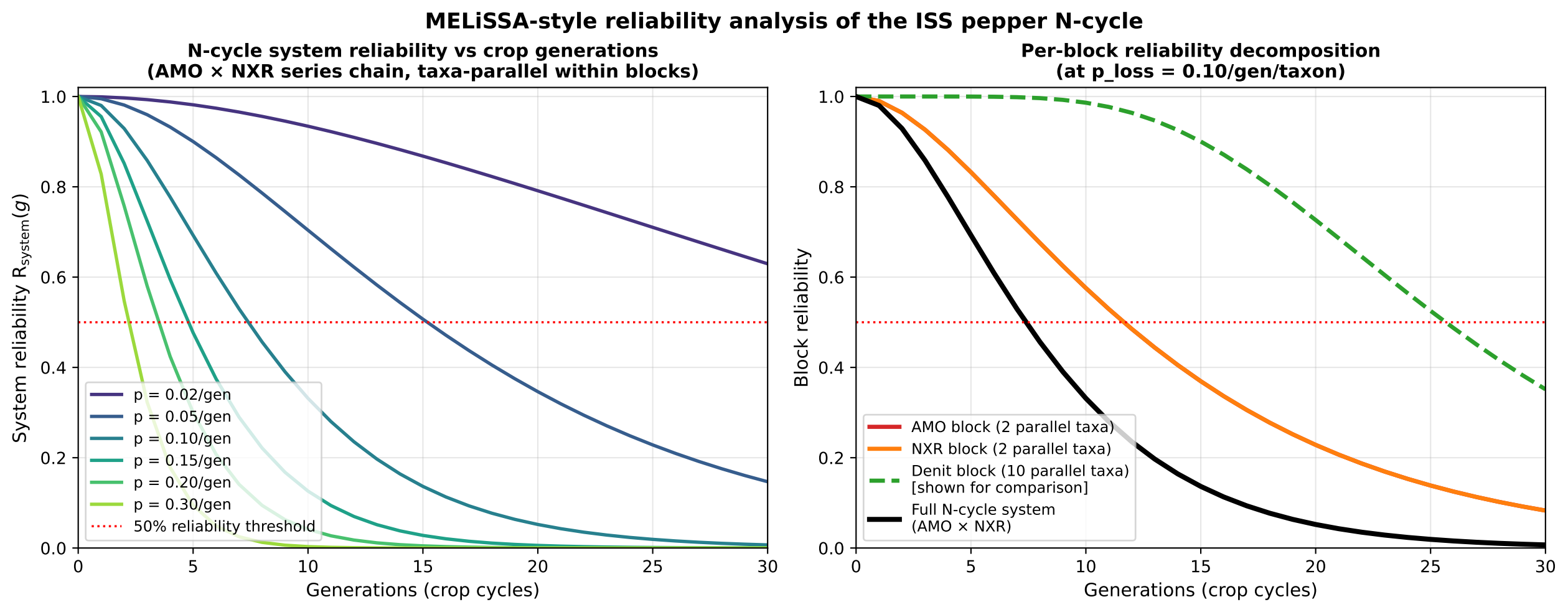
MELiSSA-style reliability analysis of the ISS pepper N-cycle
N-cycle system reliability vs crop generations
(AMO × NXR series chain, taxa-parallel within blocks)
System reliability $R_{system}(g)$
Generations (crop cycles)
p = 0.02/gen
p = 0.05/gen
p = 0.10/gen
p = 0.15/gen
p = 0.20/gen
p = 0.30/gen
50% reliability threshold
Per-block reliability decomposition
(at p_loss = 0.10/gen/taxon)
Block reliability
Generations (crop cycles)
AMO block (2 parallel taxa)
NXR block (2 parallel taxa)
Denit block (10 parallel taxa)
[shown for comparison]
Full N-cycle system
(AMO × NXR)